Minimization of Maximum Mean Discrepancy (MMD)

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About Me

- Zonghao Chen
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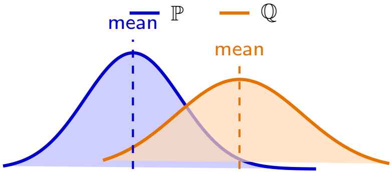
- Graduated from Tsinghua University in 2022
  - Department of EE
- Kernel (nonparametric) methods, causal inference, statistical learning theory

Content: Minimization of MMD

- What is MMD?
- Minimization of MMD for numerical integration
- Minimization of MMD for sampling / generative modelling
- Minimization of MMD for statistical inference

Background: what is MMD?

- Given two distributions $\mathbb{P}$ and $\mathbb{Q}$, how to compare them?
- Compute the difference of their means: $|\mathbb{E}_{\mathbb{P}}[X] - \mathbb{E}_{\mathbb{Q}}[X]|$.
- Compute the difference of their second moments: $|\mathbb{E}_{\mathbb{P}}[X^2] - \mathbb{E}_{\mathbb{Q}}[X^2]|$.
- The first two moments are not enough!



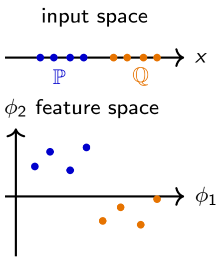
Goal: Compare $\mathbb{P}$ and $\mathbb{Q}$ with a general f, i.e., $|\mathbb{E}_{\mathbb{P}}[f(X)] - \mathbb{E}_{\mathbb{Q}}[f(X)]|$?

Background: Maximum Mean Discrepancy (MMD)

- Compare $\mathbb{P}$ and $\mathbb{Q}$ by maximizing the difference over a function class $\mathcal{H}$:

$$\mathrm{MMD}(\mathbb{P}, \mathbb{Q}) := \sup_{f \in \mathcal{H}} |\mathbb{E}_{\mathbb{P}}[f(X)] - \mathbb{E}_{\mathbb{Q}}[f(X)]|.$$

- $\mathcal{H}$ is a reproducing kernel Hilbert space (RKHS) with kernel k.
- The RKHS contains functions that are linear combinations of the feature maps $\phi : \mathcal{X} \to \mathcal{H}$, $f = \sum_{i=1}^{n} \alpha_i \phi(x_i)$.



Theorem 1

MMD admits a closed-form expression:

$$\mathrm{MMD}^2(\mathbb{P}, \mathbb{Q}) = \mathbb{E}_{X, X' \sim \mathbb{P}}[k(X, X')] + \mathbb{E}_{Y, Y' \sim \mathbb{Q}}[k(Y, Y')] - 2\mathbb{E}_{X \sim \mathbb{P},\, Y \sim \mathbb{Q}}[k(X, Y)].$$

- MMD is a probability divergence ($\mathrm{MMD}(\mathbb{P}, \mathbb{Q}) = 0 \iff \mathbb{P} = \mathbb{Q}$) if k is characteristic (Gaussian).

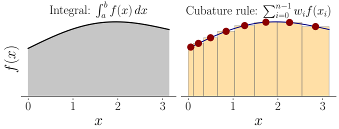
We now have MMD

A kernel-based way to compare probability distributions $\mathbb{P}$ and $\mathbb{Q}$

What can we use it for?

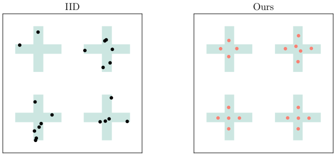
Numerical Integration

$$\mathbb{P} \stackrel{?}{=} \mathbb{Q}$$



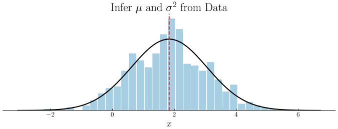
Sampling / generative modelling

$$X \sim \mathbb{Q}_\theta \rightarrow \mathbb{P}$$



Statistical inference

$$\hat{\theta}_{\mathrm{MMD}}$$



Next: MMD as both a diagnostic and an objective.